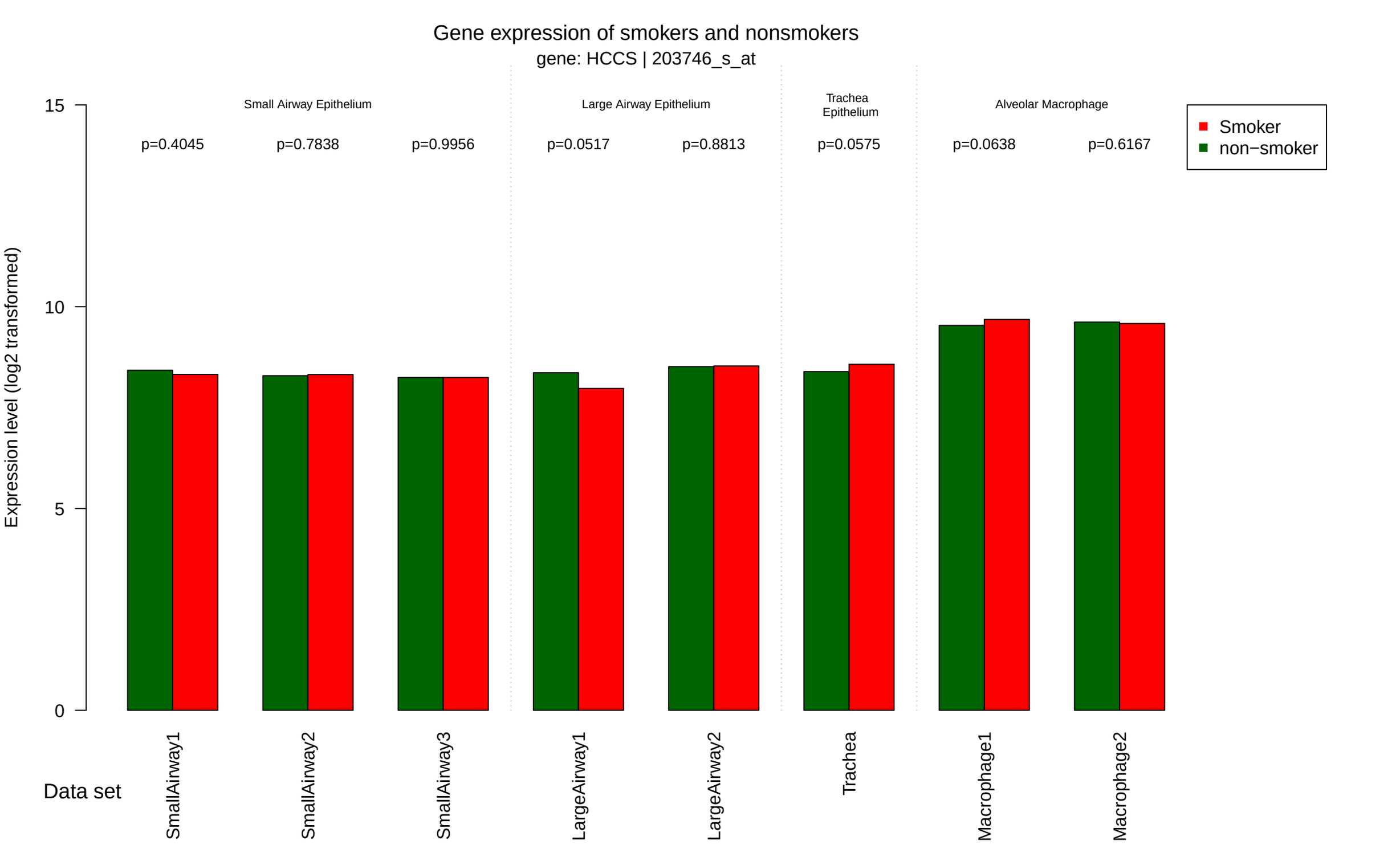
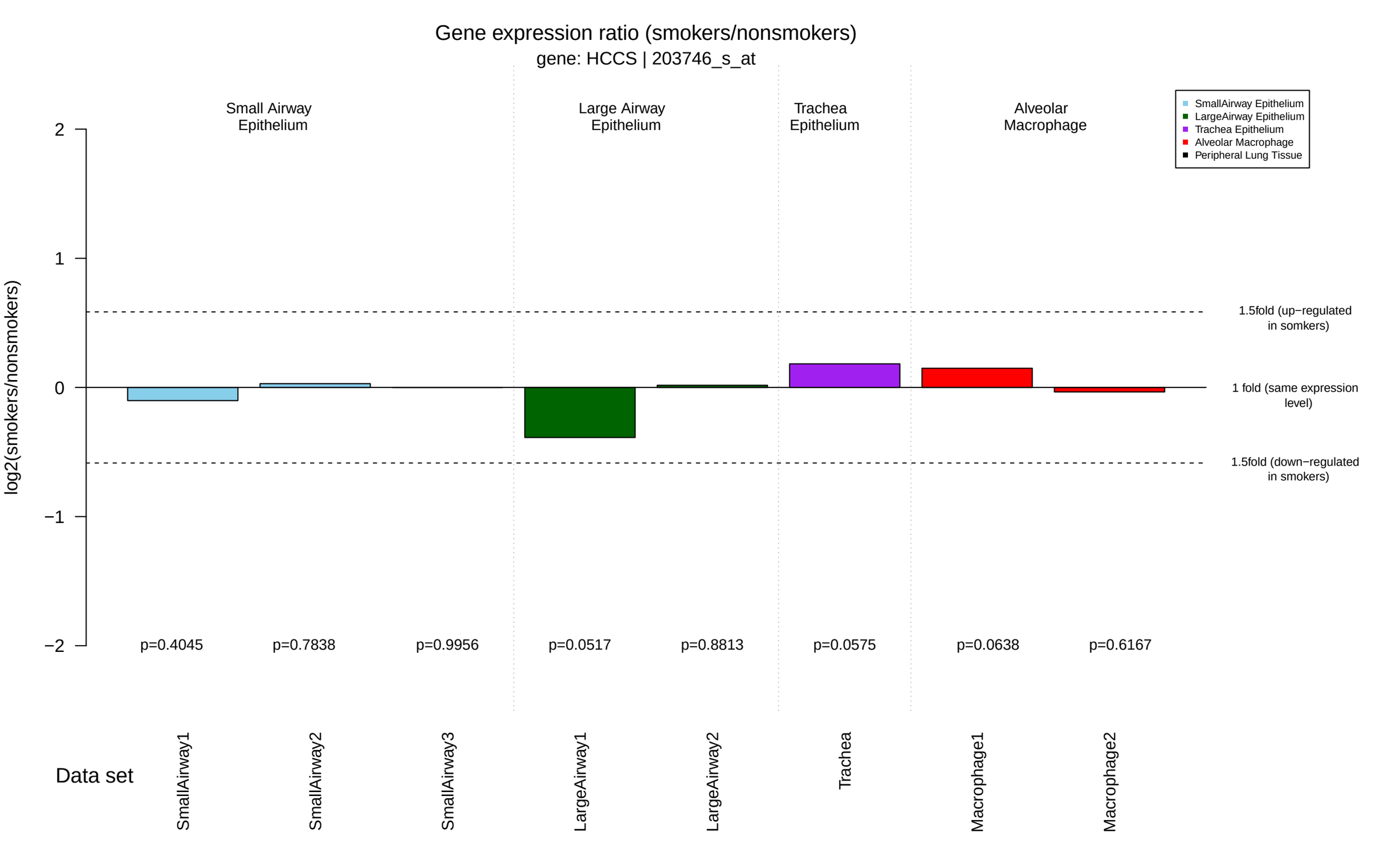
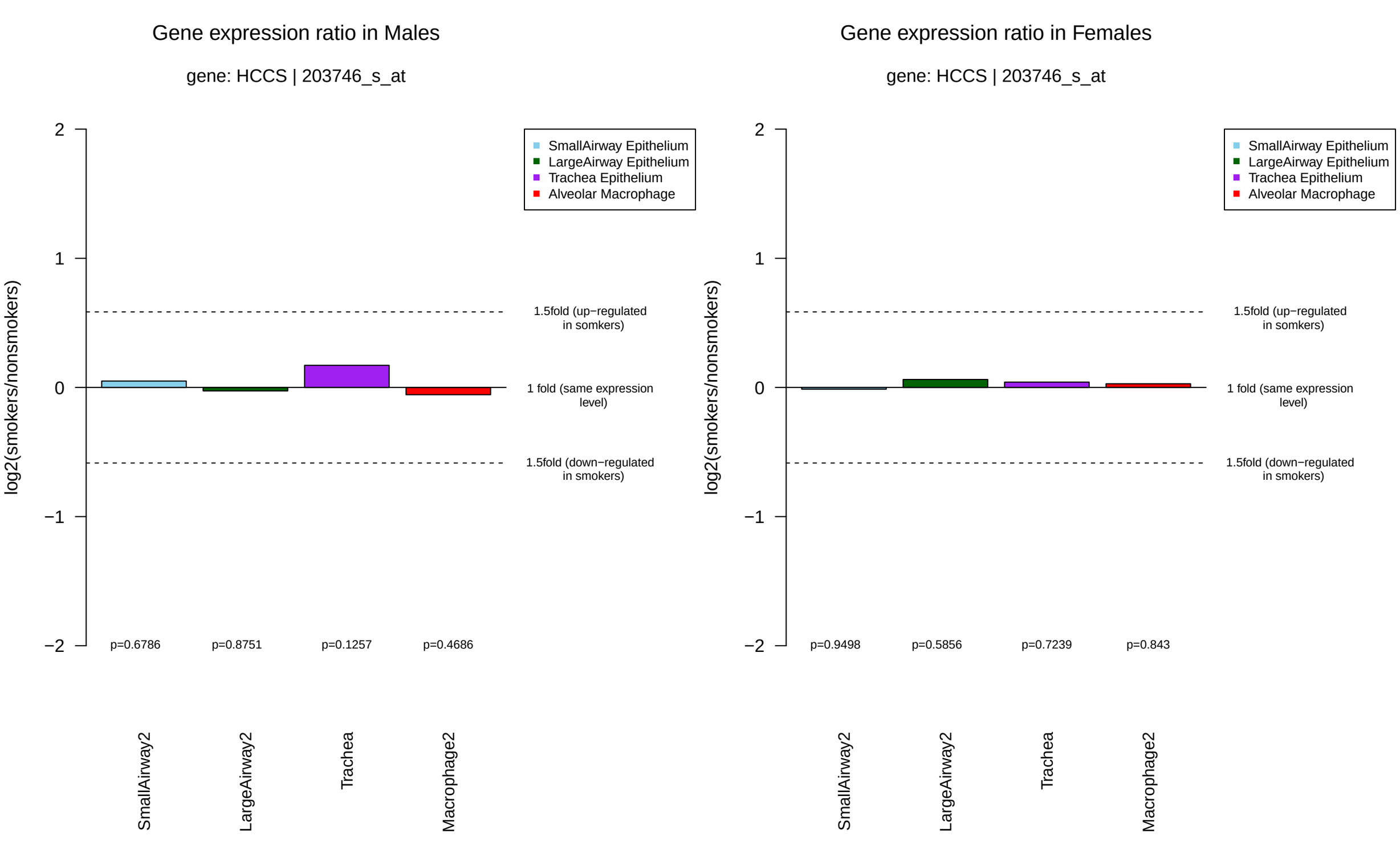
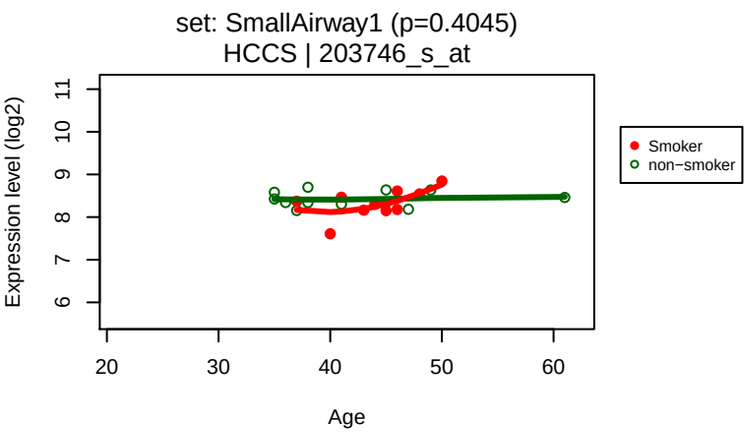
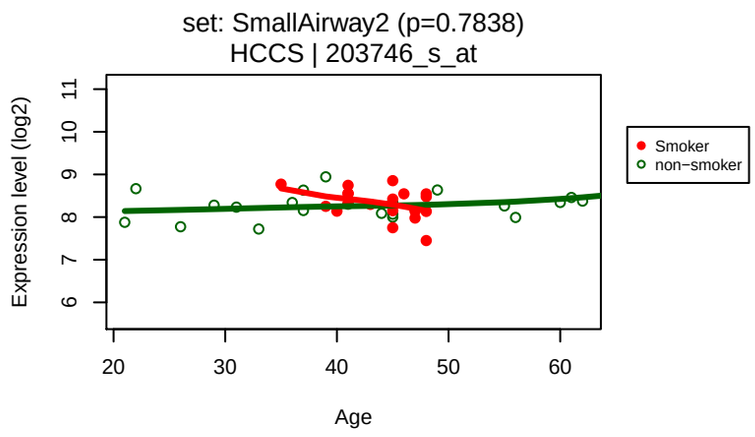
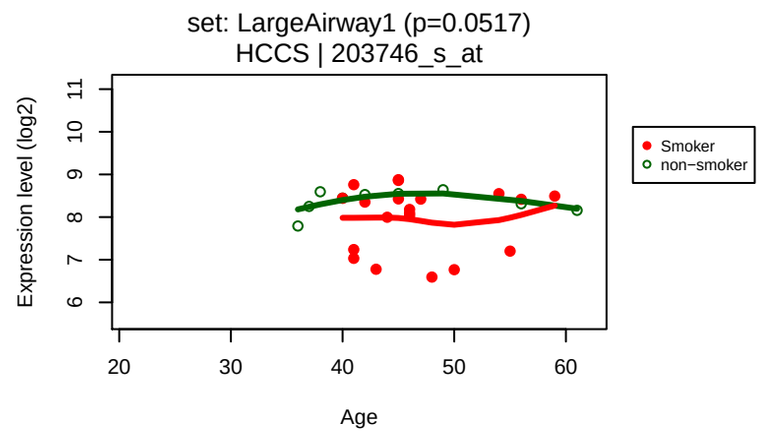

Gene expression of smokers and nonsmokers
gene: HCCS | 203746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4045
p=0.7838
p=0.9956
p=0.0517
p=0.8813
p=0.0575
p=0.0638
p=0.6167
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

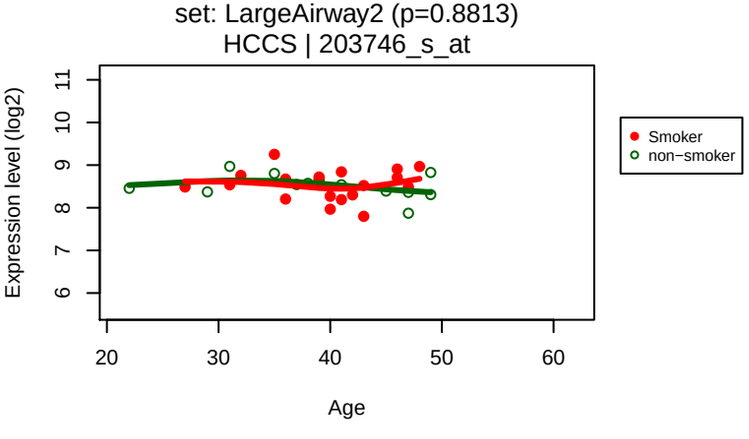

Gene expression ratio (smokers/nonsmokers)
gene: HCCS | 203746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4045
p=0.7838
p=0.9956
p=0.0517
p=0.8813
p=0.0575
p=0.0638
p=0.6167
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

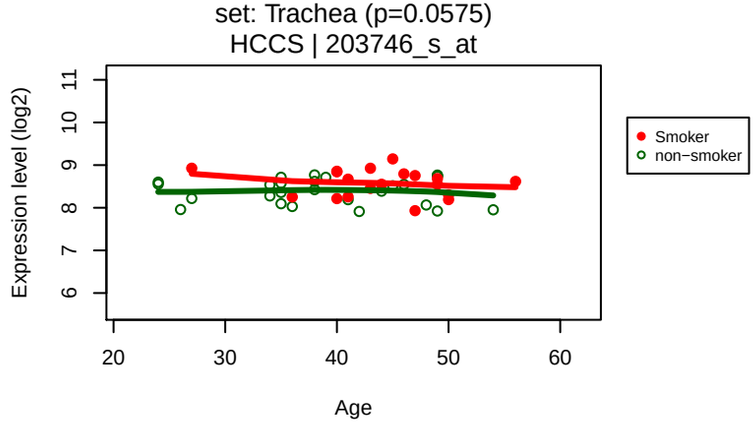

Gene expression ratio in Males
gene: HCCS | 203746_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6786
p=0.8751
p=0.1257
p=0.4686
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HCCS | 203746_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9498
p=0.5856
p=0.7239
p=0.843
SmallAirway2
LargeAirway2
Trachea
Macrophage2

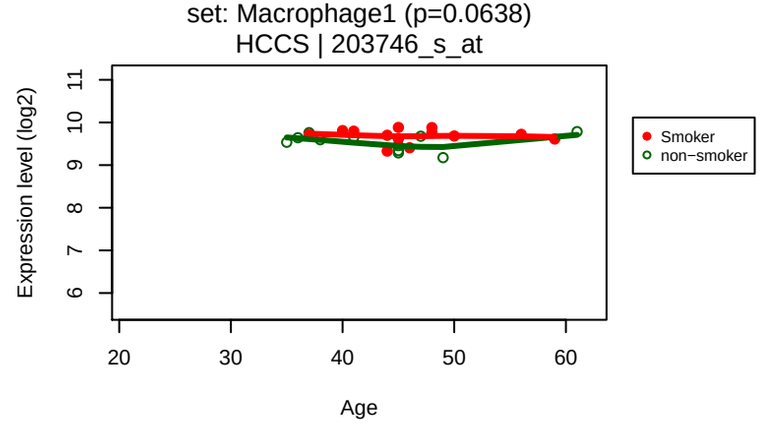

set: SmallAirway1 (p=0.4045)
HCCS | 203746_s_at
Expression level (log2)
Age
Smoker
non−smoker

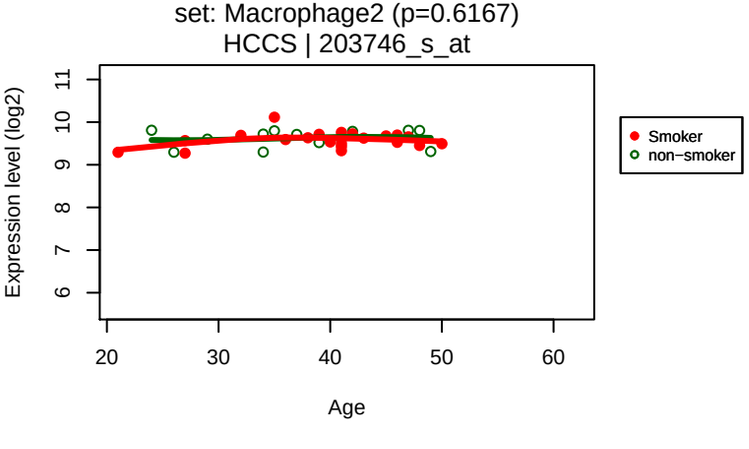

set: SmallAirway2 (p=0.7838)
HCCS | 203746_s_at
Expression level (log2)
Age
Smoker
non−smoker

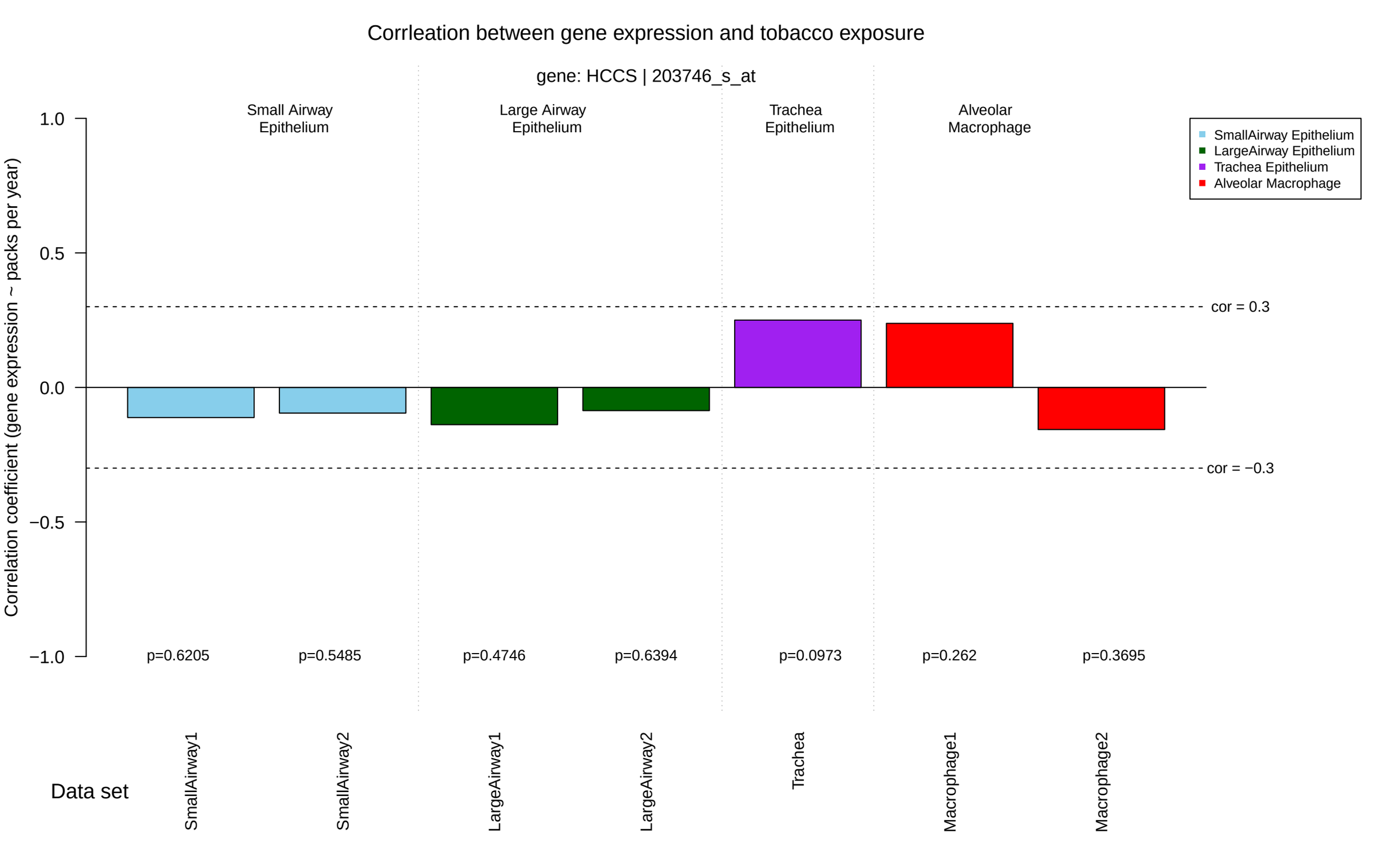

set: LargeAirway1 (p=0.0517)
HCCS | 203746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8813)
HCCS | 203746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0575)
HCCS | 203746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0638)
HCCS | 203746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6167)
HCCS | 203746_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HCCS | 203746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6205
p=0.5485
p=0.4746
p=0.6394
p=0.0973
p=0.262
p=0.3695
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2